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**Extension to SAREF;**
**Part 5: Industry and Manufacturing Domains**

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# Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Smart Machine-to-Machine communications (SmartM2M).

The present document is part 5 of a multi-part deliverable covering SmartM2M; Extension to SAREF, as identified below:

Part 1: "Energy Domain";

Part 2: "Environment Domain";

Part 3: "Building Domain";

Part 4: "Smart Cities Domain";

**Part 5: "Industry and Manufacturing Domains";**

Part 6: "Smart Agriculture and Food Chain Domain";

Part 7: "Automotive Domain";

Part 8: "eHealth/Ageing-well Domain";

Part 9: "Wearables Domain";

Part 10: "Water Domain";

Part 11: "Lift Domain";

Part 12: "Smart Grid Domain";

Part 13: "Maritime Domain".

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"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

# 1 Scope

The present document presents SAREF4INMA, a SAREF extension for the Industry and Manufacturing domains. SAREF4INMA V2.1.1 is a major revision of the SAREF4INMA, using updated reference ontology patterns [2] to solve the harmonization needs [i.12], with updated development framework and tools [3].

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 103 264 (V3.1.1): "SmartM2M; Smart Applications; Reference Ontology and oneM2M Mapping".

[2] ETSI TS 103 548: "SmartM2M; SAREF reference ontology patterns".

[3] ETSI TS 103 673: "SmartM2M; SAREF Development Framework and Workflow, Streamlining the Development of SAREF and its Extensions".

## 2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI TR 103 411 (V1.1.1) (02-2017): "SmartM2M; Smart Appliances; SAREF extension investigation".

[i.2] ETSI TR 103 507 (V1.1.1): "SmartM2M; SAREF extension investigation; Requirements for industry and manufacturing domains".

[i.3] ETSI TS 103 410-3 (V1.1.2): "SmartM2M; Extension to SAREF; Part 3: Building Domain".

[i.4] ISO/IEC 11179-6: "Information technology -- Metadata registries (MDR) -- Part 6: Registration".

[i.5] ISO 29002 (all parts): "Industrial automation systems and integration -- Exchange of characteristic data".

[i.6] ISO 6532: "Portable chain-saws -- Technical data".

[i.7] BS EN 10204 (2004): "Metallic products -- Types of inspection documents".

[i.8] IEC 61512 (all parts): "Batch control".

[i.9] ISO/IEC 11578:1996: "Information technology -- Open Systems Interconnection -- Remote Procedure Call (RPC)".

[i.10] Recommendation ITU-T X.667/ISO/IEC 9834-8:2005: "Information technology -- Procedures for the operation of object identifier registration authorities: Generation of universally unique identifiers and their use in object identifiers".

[i.11] IEC 62264 (all parts): "Enterprise-control system integration".

[i.12] ETSI TR 103 781 (V1.1.1): "SmartM2M; Study for SAREF ontology patterns and usage guidelines".

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the following terms apply:

**ontology:** formal specification of a conceptualization, used to explicitly capture the semantics of a certain reality

## 3.2 Symbols

Void.

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| BIC | Brainport Industries Campus |
| BS | British Standard |
| EAN | European Article Number |
| EN | European Norm |
| GS1 | Global Standards One |
| GTIN | Global Trade Item Number |
| GUID | Globally Unique IDentifier |
| ID | Identifier |
| IEC | International Electrotechnical Commission |
| IRDI | International Registration Data Identifier |
| ISO | International Organisation for Standardization |
| ITF | Interleaved 2 of 5 |
| ITU-T | International Telecommunication Union - Telecommunications sector |
| OWL | Web Ontology Language |
| OWL-DL | Web Ontology Language - Description Logic |
| QR | Quick Response code |
| RAMI | Reference Architectural Model Industry 4.0 |
| RFID | Radio Frequency Identification |
| RPC | Remote Procedure Call |
| SAREF | Smart Applications REFerence ontology |
| SAREF4BLDG | SAREF extension for buildings |
| SAREF4INMA | SAREF extension for industry and manufacturing domains |
| TR | Technical Report |
| TS | Technical Specification |
| UCC | Uniform Commercial Code |
| UPC | Universal Product Code |
| UPC-A | Universal Product Code |

NOTE: UPC-A is an 11 digit variation of UPC, as opposed to UPC-E which is the 6 digit variation.

UUID Universally Unique Identifier

# 4 SAREF4INMA ontology and semantics

## 4.1 Introduction and overview

The present document is a technical specification of SAREF4INMA, an extension of SAREF [1] that was created for the industry and manufacturing domain. SAREF4INMA was created to be aligned with related initiatives in the smart industry and manufacturing domain in terms of modelling and standardization, such as the Reference Architecture Model for Industry 4.0 (RAMI), which combines several standards used by the various national initiatives in Europe that support digitalization in manufacturing. These initiatives include, but are not limited to, the platform Industrie 4.0 in Germany, the Smart Industry initiative in the Netherlands, Industria 4.0 in Italy, the 'Industrie du future initiative' in France and more.

SAREF4INMA is an OWL-DL ontology that extends SAREF with 21 classes, 16 object and 10 data type properties. In addition, SAREF4INMA reuses SAREF and SAREF4BLDG ontologies by importing them. SAREF4INMA focuses on extending SAREF for the industry and manufacturing domain to solve the lack of interoperability between various types of production equipment that produce items in a factory and, once outside the factory, between different organizations in the value chain to uniquely track back the produced items to the corresponding production equipment, batches, material and precise time in which they were manufactured.

The full list of use cases, standards and requirements that guided the creation of SAREF4INMA are described in the associated ETSI TR 103 507 [i.2]. The "zero defect manufacturing" use case has been used as basis for the creation of SAREF4INMA in the present document. This use case is concerned with improving the manufacturing process in terms of flexibility to timely change from one manufactured product to another, generating as little yield loss as possible. Also the "smart services for product in use" and "smart product lifecycle" use cases are acknowledged in the associated ETSI TR 103 507 [i.2] as especially relevant for SAREF4INMA, as they pose semantic interoperability issues for, respectively:

1) the manufacturing companies that remain responsible for the proper functioning of a product during its entire lifecycle, also when the product has left the factory; and

2) the various, interacting parties involved in the value chain (e.g. manufacturer, user, servicing organization, parts supplier, etc.) that need to refer to a common digital footprint of a product to allow for its management during its entire lifecycle.

Note that SAREF4INMA specified in the present document provides a second version of the SAREF extension for the industry and manufacturing domain, based on the (limited set of) use cases mentioned above and an initial list of standards for digitalization, communication, engineering and life-cycle, covering relevant concepts such as factory, production equipment, item, material and batch, as described in ETSI TR 103 507 [i.2]. However, as all the SAREF ontologies, SAREF4INMA is a dynamic semantic model that should be used, validated and improved over time with and by the stakeholders in the industry and manufacturing domain in an iterative and interactive manner to accommodate more use cases, standards and generate new requirements as needed.

The prefixes and namespaces used in SAREF4INMA and in the present document are listed in Table 1.

**Table 1: Prefixes and namespaces used within the SAREF4INMA ontology**

| Prefix | Namespace |
|---|---|
| s4inma | https://saref.etsi.org/saref4inma/ |
| saref | https://saref.etsi.org/core/ |
| s4bldg | https://saref.etsi.org/saref4bldg/ |
| dcterms | http://purl.org/dc/terms/ |
| owl | http://www.w3.org/2002/07/owl# |
| rdf | http://www.w3.org/1999/02/22-rdf-syntax-ns# |
| rdfs | http://www.w3.org/2000/01/rdf-schema# |
| skos | http://www.w3.org/2004/02/skos/core# |
| geo | http://www.opengis.net/ont/geosparql# |
| xsd | http://www.w3.org/2001/XMLSchema# |

## 4.2 SAREF4INMA

### 4.2.1 General Overview

An overview of the SAREF4INMA ontology is provided in Figure 1, where orange rectangles are used to denote classes created in SAREF4INMA, while white rectangles denote classes reused from other ontologies, such as SAREF or SAREF4BLDG. For all the entities described in the present document, it is indicated whether they are defined in the SAREF4INMA extension or elsewhere by the prefix included before their identifier, i.e. if the element is defined in SAREF4INMA the prefix is `s4inma`, while if the element is reused from another ontology it is indicated by a prefix according to Table 1 (e.g. `saref` refers to SAREF and `s4bldg` refers to SAREF for building).

Arrows are used to represent properties between classes and to represent some RDF, RDF-S and OWL constructs, more precisely:

- Plain arrows with white triangles represent the `rdfs:subClassOf` relation between two classes. The origin of the arrow is the class to be declared as subclass of the class at the destination of the arrow.
- Dashed arrows between two classes indicate a local restriction in the origin class, i.e. that the object property can be instantiated between the classes in the origin and the destination of the arrow. The identifier of the object property is indicated within the arrow.
- Dashed arrows with no identifier are used to represent the `rdf:type` relation, indicating that the element in the origin of the arrow is an instance of the class in the destination of the arrow.

Datatype properties are denoted by rectangles attached to the classes, in an UML-oriented way. Dashed boxes represent local restrictions in the class, i.e. datatype properties that can be applied to the class they are attached to.

Individuals are denoted by rectangles in which the identifier is underlined.

Note that Figure 1 aims at showing a global overview of the main classes of SAREF4INMA and their mutual relations. More details on the different parts of Figure 1 are provided from clause 4.2.2 to clause 4.2.4.

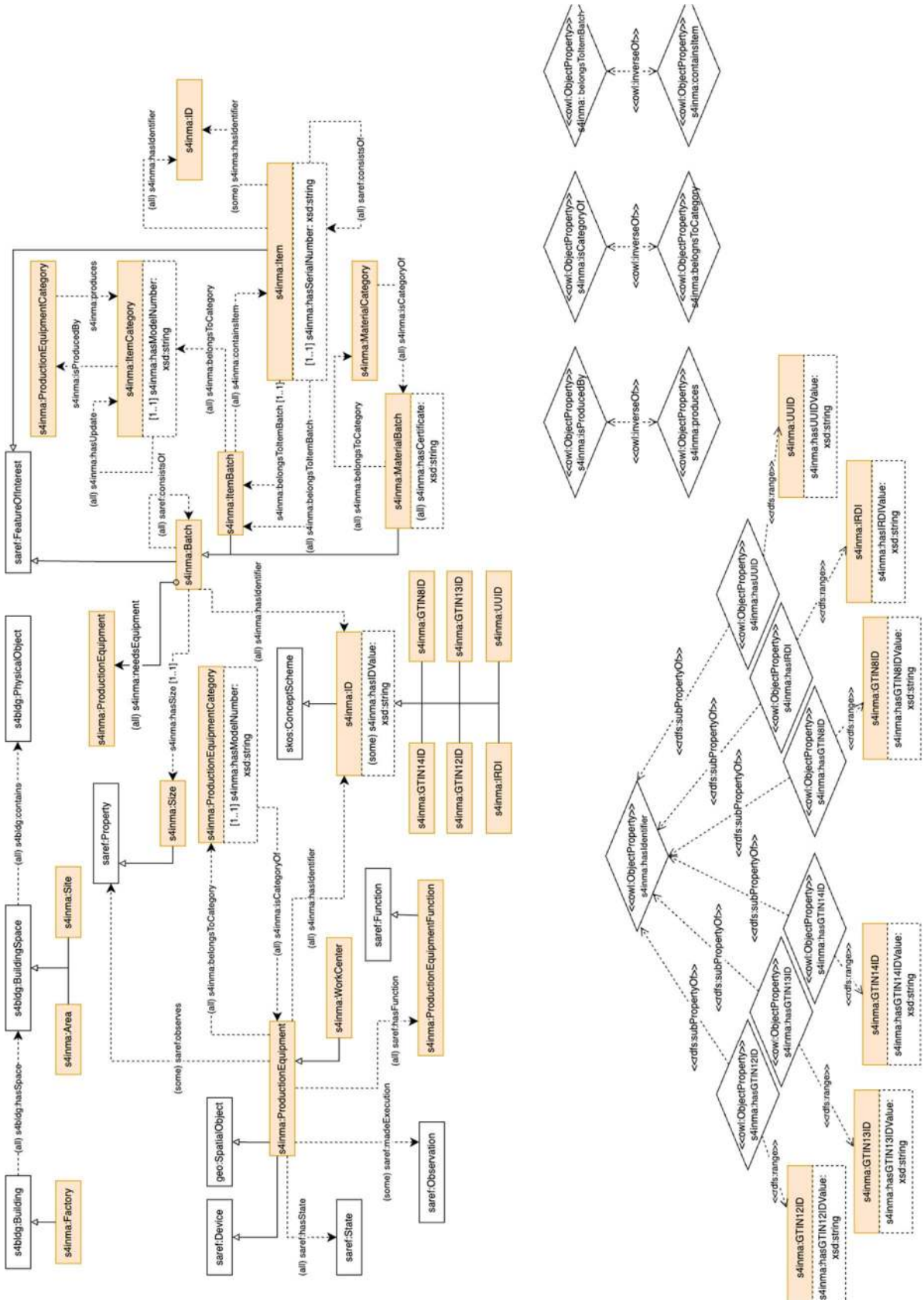



**Figure 1: SAREF4INMA overview**

### 4.2.2 Item and Batch

This clause focuses on the classes of SAREF4INMA that describe an item produced in a factory. The classes of interest, which are `s4inma:Item`, `s4inma:ItemCategory`, `s4inma:MaterialCategory`, `s4inma:Batch`, `s4inma:ItemBatch`, `s4inma:MaterialBatch` and `s4inma:ID`, are shown in Figure 3.

An Item is a tangible object that represents either the goods produced by an organization's production process or individually traced supplies (i.e. sub-assemblies of supplies). An item can be individually traced using an ID. SAREF4INMA allows to use several types of IDs, such as the Global Trade Item Number (GTIN) defined by GS1 (https://www.gs1.org/), used by organizations to uniquely identify their trade items as products or services that are priced, ordered or invoiced at any point in the supply chain. There are four GTIN formats (GTIN-8, GTIN-12, GTIN-13, GTIN-14) and SAREF4INMA defines classes and properties for each of them. SAREF4INMA defines also classes and properties to associate items to the International Registration Data Identifier (IRDI), which is based on the international standards ISO/IEC 11179-6 [i.4], ISO 29002 [i.5] and ISO 6532 [i.6]. An example of relevant standard that uses IRDIs is the eCl@ss specification (https://www.eclass.eu/en/) for grouping materials, products and services. Other types of IDs are defined in SAREF4INMA, such as the Universally Unique Identifier (UUID), or can be further defined ad-hoc by the ontology users by creating new classes as subclasses of the `s4inma:ID` class.

An Item can recursively consist of other items (e.g. a shaver consists of a shaver head, motor and body) and can be the feature of interest of a observation (e.g. a shaver can be the feature of interest of a temperature observation made by a welding machine used to join different parts in the production of the shaver). An item is created exactly in one ItemBatch, which describes a uniform collection of items produced at a certain time using a certain production equipment. An ItemBatch consists of a set of items with similar properties (e.g. a certain brand and model of sensors made using a certain production line). An ItemBatch is a specialization of the more general Batch, which can be further specialized in a MaterialBatch. The difference between ItemBatch and MaterialBatch is that individual items can be traced in an ItemBatch (e.g. it is possible to trace an individual metal sheet in an ItemBatch), whereas it is not possible to exactly trace material in a MaterialBatch, (e.g. it is not possible to trace the exact piece of raw plastic material from a MaterialBatch, as the raw plastic is a volume, not identifiable in a specific sheet like in the case of metal sheets).

Material batches can be equipped with quality certificates, such as the BS EN 10204:2004 [i.7] category 3.1 steel quality certificate (https://standardsdevelopment.bsigroup.com). These certificates provide additional information about the material in the batch. Furthermore, MaterialBatches belong to some MaterialCategory, which describes a certain type of material (e.g. a certain type of steel sheets). Analogously, item batches belong to some ItemCategory, which describes a single type of Items (e.g. a certain type of sensor). An ItemCategory is in turn produced by some ProductionEquipmentCategory (see clause 4.2.3). The essential properties of each Item in all ItemBatches are the same. However, each ItemBatch might use different MaterialBatches and/or different ProductionEquipment. Therefore, small deviations between batches might occur, while the essential properties of all Items related to an ItemCategory are similar.

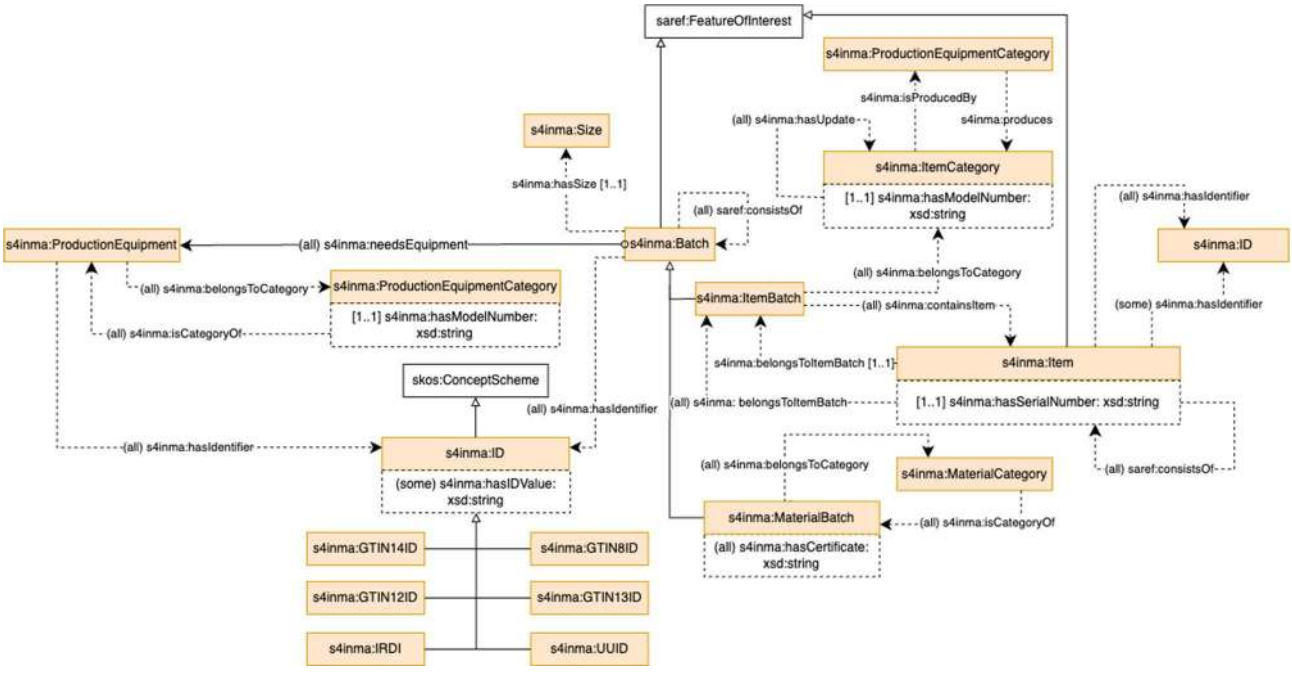



**Figure 2: Item, Batch and related classes**

Table 2 summarizes the definitions of the Item and related classes described above.

**Table 2: Item, Batch and related classes: definitions**

| Class | Definition |
|---|---|
| s4inma:Item | A tangible object which can be unique identified, for example, with a GTIN in the form of a barcode/QR/RFID tag. An item can be the result of an organization's production process (i.e. outflow of objects/goods) or a uniquely identifiable material (i.e. inflow of objects/supplies). Each item is part of exactly one ItemBatch, whereas each ItemBatch contains only Items with similar properties. An item can consist of multiple Batches and other Items (i.e. subassemblies). |
| s4inma:Batch | A uniform collection of tangible objects or Lot. This can either be a collection of produced items (i.e. the outflow of products) or a collection of raw material or required material (i.e. the inflow of products). It is assumed that the objects in a batch are similar and thus have shared attributes.<br>Note that this definition is broader than the definition in IEC 61512 [i.8], which defines a batch as the material that is being produced (whereas in SAREF4INMA a batch can be items or materials). |
| s4inma:ItemBatch | A uniform collection of tangible objects which are relevant for the production process. The ItemBatch consists of a set of objects with similar properties (e.g. a certain type of sensors or metal sheets). The difference between ItemBatch and MaterialBatch is that individual items can be traced in an ItemBatch, whereas this is not possible in a MaterialBatch, meaning that, for example, it is possible to trace the individual metal sheet used in an ItemBatch, but not the specific piece of plastic of a MaterialBatch (i.e. only the volume of plastic material from which a plastic item generated can be traced). This implies that the objects in an ItemBatch have a unique identifier (e.g. a GTIN code in the form of a barcode/QR-code or RFID tag). |
| s4inma:MaterialBatch | A uniform collection of tangible raw material which is relevant for the production process. The MaterialBatch can consist of a set of objects with similar properties (e.g. a certain type of screws) or a stock of homogeneous material (e.g. oil, water). The difference between MaterialBatch and ItemBatch is that individual items cannot be traced in a MaterialBatch, whereas this is possible in an ItemBatch, meaning that, for example, it is not possible to trace the individual screw used in a MaterialBatch. |
| s4inma:ItemCategory | An ItemCategory describes a category of item in terms of its static properties. Each ItemCategory can have multiple related ItemBatches, which all contain individual Items. The essential properties of each Item in all ItemBatches are the same. However, each batch might use different MaterialBatches and/or different ProductionEquipment. Therefore, small deviations between batches might occur, while the essential properties of all Items related to an ItemCategory are similar. |
| s4inma:MaterialCategory | A MaterialCategory describes a category of material in terms of its static properties. Examples are: a certain category of steel or plastic. Each MaterialCategory can have multiple related MaterialBatches, which represent the physical material. The essential properties of the material in all MaterialBatches are the same. However, each batch might use different MaterialBatches and/or different ProductionEquipment. Therefore, small deviations between batches might occur, while the essential properties of the material related to an MaterialCategory are similar. |
| s4inma:ID | A unique identifier. |
| s4inma:GTIN8ID | GTIN-8 (EAN/UCC-8) is an 8-digit number used predominately outside of North America. |
| s4inma:GTIN12ID | GTIN-12 (UPC-A) is a 12-digit number used primarily in North America. |
| s4inma:GTIN13ID | GTIN-13 (EAN/UCC-13) is a 13-digit number used predominately outside of North America. |
| s4inma:GTIN14ID | GTIN-14 (EAN/UCC-14 or ITF-14 or also known as ITF Symbol, SCC-14, DUN-14, UPC Case Code, UPC Shipping Container Code, UCC Code 128, EAN Code 128) is a 14-digit number used to identify trade items at various packaging levels. |
| s4inma:IRDI | International Registration Data Identifier (IRDI) is based on the international standards ISO/IEC 11179-6 [i.4], ISO 29002 [i.5] and ISO 6532 [i.6] and used in eCl@ss and the Asset Administration Shell as unique identifier. |
| s4inma:UUID | A universally unique identifier (UUID) is a 128-bit number used to identify items and is also known as: globally unique identifier (GUID). In its canonical textual representation, the sixteen octets of a UUID are represented as 32 hexadecimal (base 16) digits, displayed in five groups separated by hyphens, in the form 8-4-4-4-12 for a total of 36 characters (32 alphanumeric characters and four hyphens). UUID are documented in ISO/IEC 11578 [i.9] and in Recommendation ITU-T X.667 \| ISO/IEC 9834-8:2005 [i.10]. |
| s4inma:Size | The amount of certain objects in a collection (e.g. size of a material batch). |

Table 3 summarizes the properties that characterize an Item and the related classes described above.

**Table 3: Properties of Item, Batch and related classes**

| Property | Definition |
|---|---|
| s4inma:Item s4inma:belongsToItemBatch **only** s4inma:ItemBatch | The relation between an Item and the ItemBatch in which it is created (inverse of s4inma:containsItem). |
| s4inma:Item s4inma:belongsToItemBatch **exactly** 1 s4inma:ItemBatch | An Item is created exactly in one ItemBatch. |
| s4inma:Item saref:consistsOf **only** s4inma:Item | An item can recursively consist of other Items. |
| s4inma:Item s4inma:hasSerialNumber **max** 1 xsd:string | An Item can have one serial number at most. |
| s4inma:Item s4inma:hasIdentifier **only** s4inma:ID | The relation between an item and its unique identifier. |
| s4inma:Item s4inma:hasIdentifier **some** s4inma:ID | An Item has a unique identifier (e.g. GTIN, IRDI, UUID, etc.). |
| s4inma:ItemCategory s4inma:hasIdentifier **only** s4inma:ID | The relation between an item category and its unique identifier. |
| s4inma:ItemCategory s4inma:hasIdentifier **some** s4inma:ID | An item category has a unique identifier (e.g. GTIN, IRDI, UUID, etc.). |
| s4inma:ItemCategory s4inma:hasModelNumber **max** 1 xsd:string | An ItemCategory can have one model number at most. |
| s4inma:ItemCategory saref:hasManufacturer **max 1** xsd:string | A relation from SAREF identifying the manufacturer of an entity . |
| s4inma:ItemCategory s4inma:hasUpdate **only** s4inma:ItemCategory | An ItemCategory can have a new revision. The old version is related to the new version via the hasUpdate relation. |
| s4inma:ItemCategory saref:hasVersion **max** 1 xsd:string | An ItemCategory can have one version number at most. The combination of hasModelNumber and hasVersion should be unique. |
| s4inma:ItemCategory s4inma:isCategoryOf **only** s4inma:ItemBatch | A relation between a certain category of items and the associated item batches. |
| s4inma:ItemCategory s4inma:isProducedBy **some**s4inma:ProductionEquipmentCategory | An ItemCategory is produced using certain categories of machines (i.e. ProductionEquipmentCategory). |
| s4inma:Batch saref:consistsOf **only** s4inma:Batch | A Batch can recursively consist of other batches. |
| s4inma:Batch s4inma:hasIdentifier **only** s4inma:ID | A Batch can have a unique identifier. |
| s4inma:Batch s4inma:hasSize **max** 1 Size | A relation to count the amount of certain objects in a collection (e.g. size of a batch). |
| s4inma:Batch s4inma:needsEquipment **only** s4inma:ProductionEquipment | A relation indicating that an entity needs a particular equipment to be produced. |
| s4inma:ItemBatch s4inma:belongsToCategory **only** s4inma:ItemCategory | An ItemBatch belongs to a certain category of Items. There can be multiple batches per category (i.e. type) of items. |
| s4inma:ItemBatch s4inma:containsItem **only** s4inma:Item | A relation of a production process (e.g. Batch) that creates one or more tangible objects (e.g. Items). Inverse of s4inma:belongsToItemBatch. |
| s4inma:MaterialBatch s4inma:belongsToCategory **exactly** 1 s4inma:MaterialCategory | A MaterialBatch belongs to a certain category of Material. There can be multiple batches per category. |
| s4inma:MaterialBatch s4inma:hasCertificate **only** xsd:string | The MaterialBatch can contain material quality certificates, for example a BS 10204:2004 [i.7] clause 3.1 steel certificate. |

### 4.2.3 Production Equipment and Factory

This clause focuses on the classes that describe how a production equipment is organized and how it can exchange information within the factory. The classes of interest are: `s4inma:ProductionEquipment`, `s4inma:ProductionEquipmentCategory`, `s4inma:WorkCenter`, `s4inma:Area`, `s4inma:Site`, and `s4inma:Factory`, and are shown in Figure 3.

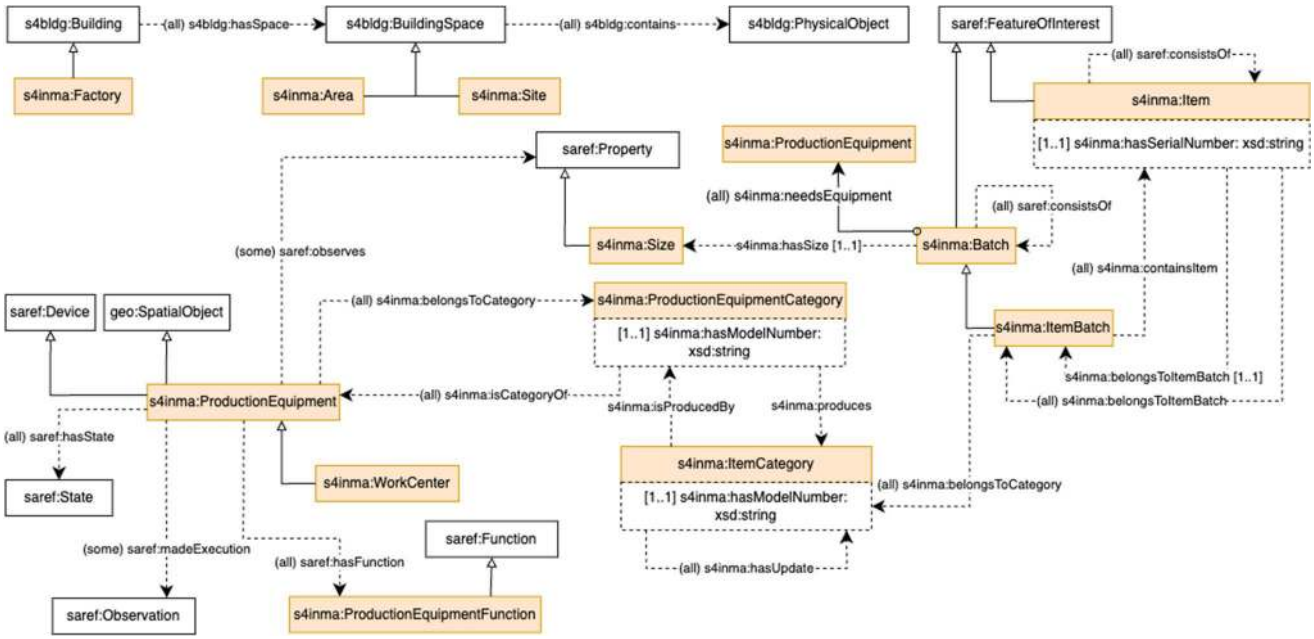



**Figure 3: Production Equipment, Factory and related classes**

A ProductionEquipmentCategory describes the kind of production equipment required for producing a certain item, i.e. a category of machine. An organization might have multiple instances of the same category of machines. Each individual machine is represented by a ProductionEquipment, which is a subclass of `saref:Device`, which is in turn a subclass of `s4bldg:PhysicalObject`. The latter is part of the SAREF for Building extension [i.3], which defines the `saref:Device` class as a subclass of the more general `s4bldg:PhysicalObject` class, following a pattern that allows to locate devices within the building. Analogously, SAREF4INMA reuses the same pattern to locate a production equipment in the factory.

In order to locate the ProductionEquipment, a factory layout can be created. A factory is represented by the `s4inma:Factory` class (which is subclass of the `s4bldg:Building` class) and can be further divided into smaller spaces using the `s4bldg:BuildingSpace` class. For the scope of SAREF4INMA, two types of BuildingSpaces are defined, namely Site and Area. A Factory can be further divided in sites, which according to IEC 62264 [i.11] are identified physical, geographical, and/or logical component groupings of a manufacturing enterprise. A Site can be divided in areas which are defined by IEC 62264 [i.11] as physical, geographical or logical groupings of resources determined by the site. An Area contains one or multiple work centers, which are a subclass of the `s4inma:ProductionEquipment` class and are defined according to IEC 62264 [i.11] as equipment elements under an area in a role-based equipment hierarchy that performs production, storage or material movement.

Note that as a subclass of `saref:Device`, a production equipment in SAREF4INMA inherits all the properties of devices defined in SAREF. This includes the possibility to associate a device (and therefore a production equipment) with a number of functions. For the purpose of SAREF4INMA, a new class of functions is created, namely the `s4inma:ProductionEquipmentFunction` class, which can be populated with subclasses that describe relevant functions, depending on the use case under consideration.

Table 4 summarizes the properties that characterize the Production Equipment and Factory classes described above.

**Table 4: Production Equipment, Factory and related classes: definitions**

| Class | Definition |
|---|---|
| s4inma:ProductionEquipment | A production equipment is a specialization of a saref:Device and s4bldg:PhysicalObject that can produce items in a manufacturing process. This class represents an individual production equipment machine and includes their specification in terms of functions, states and services. Different types of machines can be defined under this class as needed, for example, LaserCuttingMachine (i.e. a type of production equipment to cut steel material), MillingMachine (i.e. to drill holes in steel material), MouldingMachine (i.e. to mould liquid material, such as iron or plastic, and let it harden in a certain shape), WeldingMachine (i.e. to join together parts of material, such as steel), etc. |
| s4inma:ProductionEquipmentCategory | A ProductionEquipmentCategory represents a certain category of production equipment in terms of its static properties (e.g. a certain model and brand). Each ProductionEquipmentCategory can have multiple related ProductionEquipment, which represent the actual individual machines. Moreover, each ItemCategory can be produced by multiple ProductionEquipmentCategories. |
| s4inma:Factory | A subclass of s4bldg:Building specialized for the purpose of SAREF4INMA, a factory represents one or more organizations sharing a definite mission, goals and objectives which provides an output such as a product (definition taken from IEC 62264 [i.11]). A factory can be divided in one or multiple sites. |
| s4inma:Site | A subclass of s4bldg:BuildingSpace used to define the physical spaces of the building. According to IEC 62264 [i.11], sites are identified physical, geographical, and/or logical component groupings of a manufacturing enterprise. A factory can be divided in sites, whereas sites can be divided areas. |
| s4inma:Area | A subclass of s4bldg:BuildingSpace used to define the physical spaces of the building. According to IEC 62264 [i.11], areas are physical, geographical or logical groupings of resources determined by the site. A site can be divided in areas, whereas areas contain work centers. |
| s4inma:WorkCenter | A subclass of s4inma:ProductionEquipment (and therefore of s4bld:PhysicalObject). It is an equipment element under an area in a role-based equipment hierarchy that performs production, storage or material movement (definition taken from IEC 62264 [i.11]). An Area contains work centers. |

Table 5 summarizes the properties that characterize a Production Equipment and the related classes described above.

**Table 5: Properties of Production Equipment and Production Equipment Category**

| Property | Definition |
|---|---|
| s4inma:ProductionEquipmentCategory saref:hasName **only** xsd:string | A ProductionEquipmentCategory can be described using a name. |
| s4inma:ProductionEquipmentCategory s4inma:hasModelNumber **only** xsd:string | A ProductionEquipmentCategory can be described using a model number. |
| s4inma:ProductionEquipmentCategory saref:hasManufacturer **only** xsd:string | A ProductionEquipmentCategory can be described using the manufacturer of the machine. |
| s4inma:ProductionEquipment s4inma:belongsToCategory **only** s4inma:ProductionEquipmentCategory | ProductionEquipment belongs to a certain category. There can be multiple individual ProductionEquipment per category. |
| s4inma:ProductionEquipment s4inma:hasIdentifier **only** s4inma:ID | The relation between a ProductionEquipment and its unique identifier. |
| s4inma:ProductionEquipment saref:hasManufacturer **max 1** xsd:string | A relation inherited from SAREF used in SAREF4INMA to identify the manufacturer of a production equipment. |
| s4inma:ProductionEquipment saref:haModel **max 1** xsd:string | A relation inherited from SAREF used in SAREF4INMA to identify the model of a production equipment. |
| s4inma:ProductionEquipment saref:consists of **only** saref:Device | A relation inherited from SAREF used in SAREF4INMA to indicate a composite production equipment that can recursively consist of other devices (e.g. sensors and actuators). |
| s4inma:ProductionEquipment saref:hasFunction **min 1** s4inma: ProductionEquipmentFunction | A relation inherited from SAREF used in SAREF4INMA to identify the type of function of a production equipment. |
| s4inma:ProductionEquipment saref:hasState **only** saref:State | A relation inherited from SAREF used in SAREF4INMA to identify the type of state of a production equipment. |
| s4inma:ProductionEquipment saref:observes **some** saref:Property | A relation inherited from SAREF used in SAREF4INMA to specify the Property that can be measured by a certain production equipment (or other devices composing it, such as sensors and actuators). |
| s4inma:ProductionEquipment saref:madeExecution **some** saref:Observation | A relation inherited from SAREF used in SAREF4INMA between a production equipment or other devices composing it (e.g. sensors and actuators) and the observations they make. |

### 4.2.4 Observation

An important aspect of SAREF4INMA is the ability to trace back production process observations to individual items or batches. The modelling of observations in SAREF4INMA totally relies on the observations model proposed in SAREF. This modelling include the `saref:FeatureOfInterest` class that provides the means to refer to the real world phenomena that is being observed in the given observation (e.g. a shaver is an item resulting from a certain production process and it can be defined as the feature of interest of a temperature observation made by a welding machine used to join different parts in the production of the shaver). The reader shall refer to the SAREF specification for details about the modelling of observations. The following properties are reused in SAREF4INMA to complete the model of observations:

- `saref:isPropertyOf` (and its inverse `saref:hasProperty`) to link the property being observed with the feature of interest.
- `saref:observes` (and its inverse `saref:isObservedBy`) to link a given observation with the feature of interest being observed.
- `saref:madeBy` has been included as complement of the `saref:madeExecution` as its inverse, to link a observation and the device that produces it.

Within the SAREF4INMA use cases there is a need to distinguish between actual measurements and expected ones. For the former could be represented by stating a property value for a `saref:FeatureKind`, while the latter could be stated stating the value of a `saref:FeatureOfInterest` that related to the `saref:FeatureKind`, or by stating an `saref:Observation`. This enables the calculation of deviations between planned and actual production process observations.

As a `saref:Device` can recursively consist of devices, a ProductionEquipment in SAREF4INMA can also consist of other devices, such sensors and actuators. A device (e.g. production equipment and its sensors) can make observations. These observation scan be related to a specific `s4inma:Batch` or `s4inma:Item` (which are both subclasses of the `saref:FeatureOfInterest` class) via the `hasFeatureOfInterest` relation. Moreover, according to the observations model in SAREF, observations are related to the property they observe (e.g. welding temperature) and its unit of measure (e.g. degrees Celsius).

## 4.3 Instantiating SAREF4INMA

This example instantiation is referred to using the `ex` prefix. This prefix is different from the `s4inma` prefix, which indicates the SAREF4INMA ontology on which the `ex` example instantiation is built upon.

The example is shown in Figure 4 and represents an instance of a shaver (i.e. the `ex:Shaver10023`) of the `s4inma:Item` class, which is an item created in a certain batch (represented by the `ex:PhilBrau_S40_Premium_Gold_Shaver_ItemBatch392` instance), which in turn belongs to a category of items called *PhilBrau S40 Premium Gold Shaver ItemCategory*. This item category is represented by the `ex:PhilBrau_S40_Premium_Gold_Shaver _ItemCategory` instance of the `s4inma:ItemCategory` class, it has model number `ex:nr98647656` and manufacturer `PhilBrau`, and is produced using a certain production equipment category, namely the `ex:Lazor_Series_8030_ProdEquipCategory` instance of the `s4inma:ProductionEquipmentCategory` class.

The `ex:Lazor_Series_8030_ProdEquipCategory` instance is the general category of a specific production equipment, namely the `ex:Laser_Cutting_Machine_1` instance of a laser cutting machine created specifically for this example (i.e. the `ex:LaserCuttingMachine` class created in this example as a subclass of the `s4inma:ProductionEquipment` class).

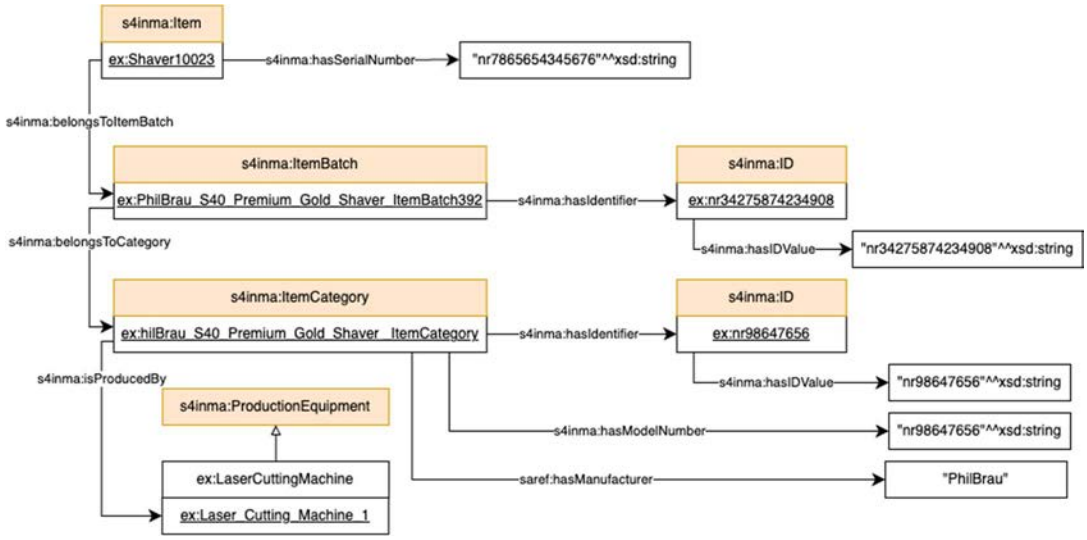



**Figure 4: Item example**

Figure 5 further shows that the *Shaver10023* item recursively consists of other three items, namely the *ShaverHead3002*, *StepMotor083* and *ShaverBody9440* items. In other words, SAREF4INMA allows to describe an item as a whole (i.e. the shaver) or in its parts (i.e. the shaver head, motor and body). The *ShaverBody9440* item is created in the *PhilBrau_S40-S50_Generic_Body_ItemBatch3290* item batch, which in turn consists of material from other batches, namely the *Torx screws_D2mm_L8mm MaterialBatch323* and *ABS_Plastic_Role_8mm_MaterialBatch742*. These material batches belong to two different material categories classes created specifically for this example, respectively the `ex:Screw` class (with its `ex:Torx_screws_D2mm_L8mm_MaterialCategory` instance) and the `ex:Plastic` class (with its `ex:ABS_Plastic_Role_8mm_MaterialCategory` instance), which are both subclasses of `s4inma:MaterialCategory`. In other words, the body of a shaver is an item created in a batch that is made of other materials such as screws and plastic.

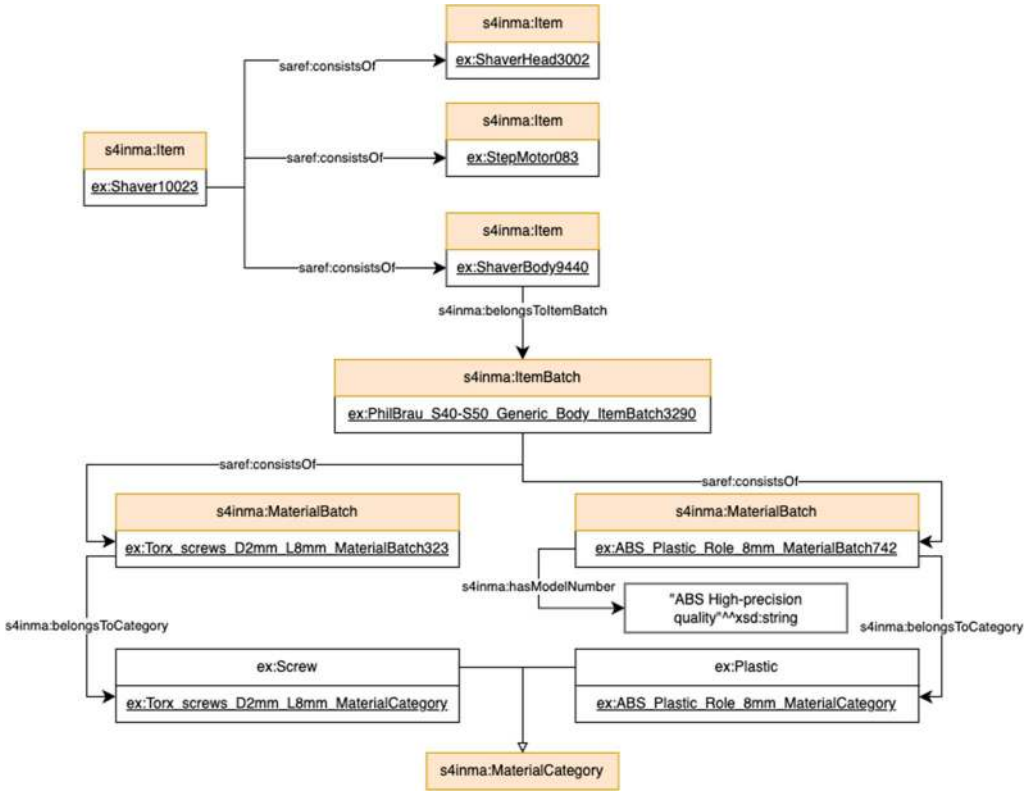



**Figure 5: Material example**

The example instantiation further defines two types of production equipment categories, namely the *Lazor_Series_8030_ProdEquipCategory* and the *WandI_Welding_Series_1000_ProdEquipCategory*. These categories represent a certain model of production equipment and not the individual machines, since an organization might have multiple machines of the same model. In particular, there is one laser cutting machine of type *Lazor_Series_8030_ProdEquipCategory*, namely the *Laser_Cutting_Machine_1*, and two welding machines, namely *Welding_Machine*_1 and *Welding_Machine_2*, which are shown in Figure 6.

These machines are instances of the `ex:WeldingMachine` and the `ex:LaserCuttingMachine` classes created for this example, which are both subclasses of the `s4inma:ProductionEquipment` class, which is in turn a subclass of `saref:Device`, which is in turn a subclass of `s4bldg:PhysicalObject`. The subclass relation of `saref:Device` ensures that a `s4inma:ProductionEquipment` can reuse SAREF functionality by inheritance, such as the possibility to perform functions, be composed by other devices such as sensors (e.g. temperature sensors), control properties (e.g. welding temperature) and make observations. For example, the *Welding_Machine_2* production equipment can perform a *JoiningFunction* (`ex:JoiningFunction` instance), controls the *WeldingTemperature* property, and further consists of the *WeldingMachineTemperatureSensor1*.

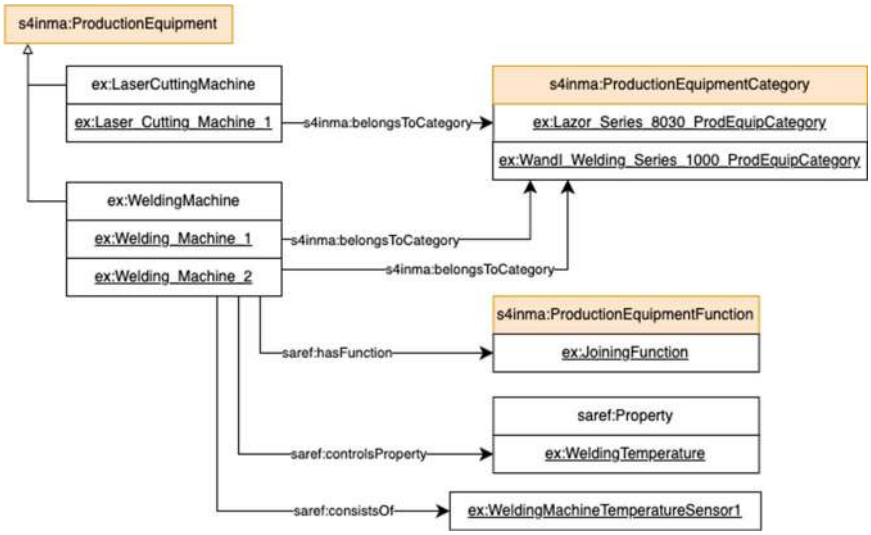



**Figure 6: Production Equipment example**

The *Welding Machine Temperature Sensor 1* makes some temperature observation during the production of *Shaver10023* in intervals of ten seconds. Figure 7 shows some example observations related to the production of the *Shaver10023* item and the reuse of the SAREF model for observations. For example, the `ex:Welding_Machine_2_Observation_W101520` instance is observed by the *WeldingMachineTemperatureSensor1*, relates to the *WeldingTemperature* property, has value *223* and unit of measure *degree Celsius*, has timestamp *2019-01-28T12:11:10* and has the item *Shaver10023* as feature of interest.

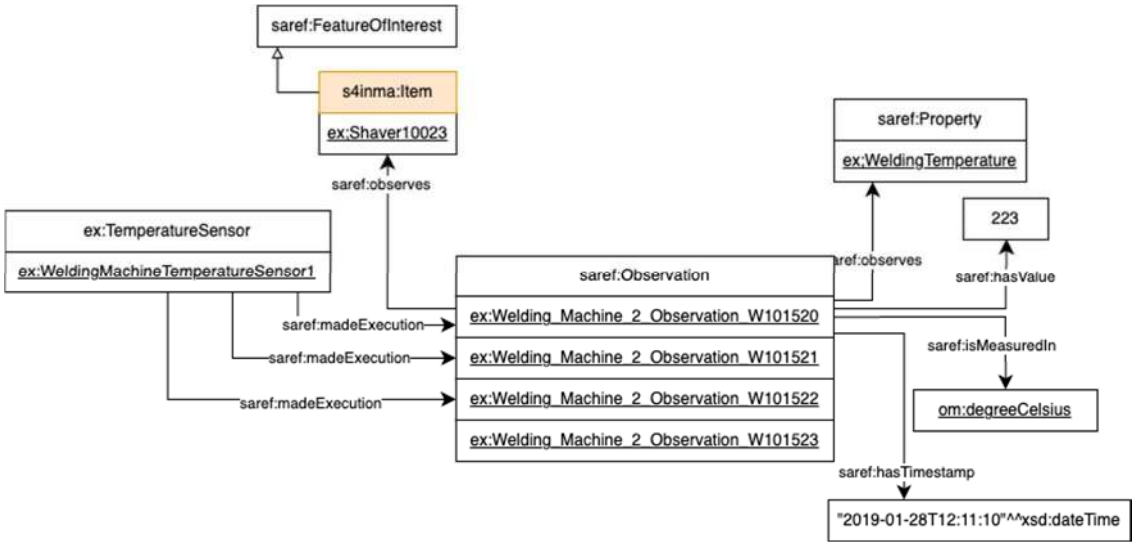



**Figure 7: Observation example**

Since a `s4inma:ProductionEquipment` is a subclass of a `saref:Device` and consequently of `s4bldg:PhysicalObject`, it is possible to assign each production equipment instance to a physical location within the factory. Figure 8 shows an instance of a `s4inma:Factory` class, which in turn is defined in SAREF4INMA as a `s4bldg:Building` subclass. This instance (`ex:Eindhoven_BIC`) represents a factory that can be decomposed into *Site* and *Area*, building spaces, which are all subclasses of `s4bldg:BuildingSpaces`. Moreover, the welding machines and the laser cutting machine are part of the *Welding_WorkCenter*, which is a *WorkCenter* located in the *Area BIC_Site_A_Area_19*, which is in its turn located in the *Site BIC_Site_A* in the *Eindhoven BIC* building.

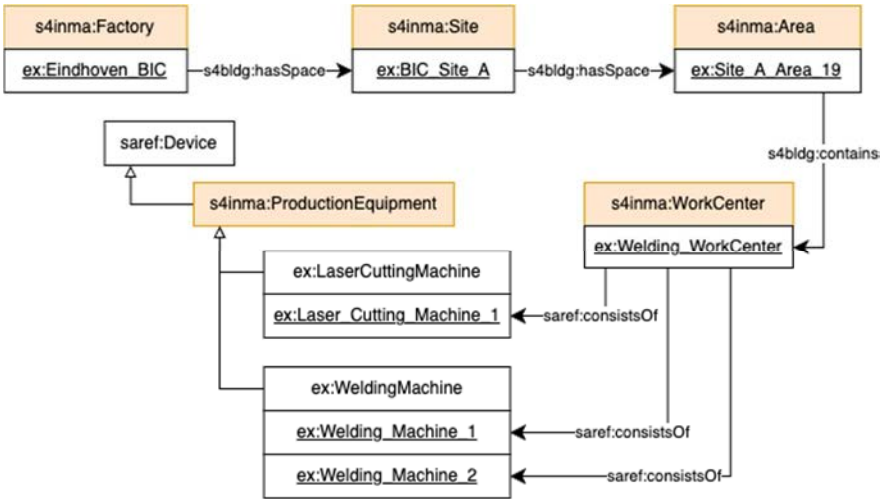



**Figure 8: Factory example**

# Annex A (informative): Bibliography

- ETSI TS 103 267: "SmartM2M; Smart Appliances; Communication Framework".
- ETSI TS 102 689: "Machine-to-Machine communications (M2M); M2M Service Requirements".

# History

| Document history | | |
|---|---|---|
| V1.1.1 | May 2019 | Publication |
| V1.1.2 | May 2020 | Publication |
| V2.1.1 | July 2024 | Publication |
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